WORKED SOLUTION

STEP III 1991, Question 4

Pure • Lines and Planes in 3D

STEP III 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q4

The key idea

Two different lines of sight from the same observer O_1 both pass through O_1, so they span a plane, and the path of P meets both of them. A line that meets two lines through O_1, without passing through O_1 itself, must lie in their plane. The same goes for O_2, so the path is the line where two planes meet. That turns a problem about four skew-looking lines into a cross product and a pair of simultaneous equations.

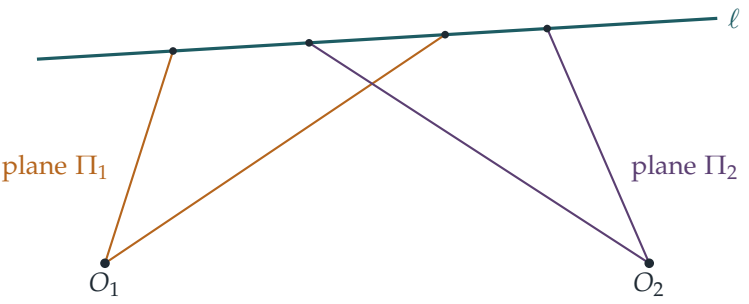
Solution

Write ℓ for the line on which P moves. The four lines of sight have directions

$$\mathbf{d}_1 = (2,0,3), \qquad \mathbf{d}_1' = (0,3,4) \qquad \text{(through } O_1\text{)},$$

$$\mathbf{d}_2 = (-3,1,3), \qquad \mathbf{d}_2' = (-4,2,5) \qquad \text{(through } O_2\text{)}.$$

Each line of sight joins an observer to a position of P, so ℓ meets all four of them.



(A schematic only: the two amber sight lines and ℓ lie in one plane Π_1, the two plum ones and ℓ in another plane Π_2.)

Why ℓ lies in a plane through each observer

First, ℓ cannot pass through O_1. If it did, then whenever $P \neq O_1$ the line of sight from O_1 would be ℓ itself, so there could not be two different lines of sight from O_1. Likewise ℓ does not pass through O_2.

The two sight lines from O_1 are different lines through O_1, so they meet only at O_1, and they span a plane Π_1. The line ℓ meets them at two points, and these are different, since the only common point is O_1 and ℓ misses O_1. Both points lie in Π_1, so the whole of ℓ lies in Π_1. In the same way ℓ lies in the plane Π_2 through O_2 containing $\mathbf{d}_2$ and $\mathbf{d}_2'$.

> **Ask yourself** Couldn't ℓ just cross each sight line at a point, without lying in any particular plane?

For a single line, yes. But here two of the crossing points are on lines through O_1, and two distinct points determine ℓ. Both points are in Π_1, so ℓ has no choice: it is the line through them, which lies in Π_1.

The plane Π_1

A normal to Π_1 is

$$\mathbf{d}_1 \times \mathbf{d}_1' = (2,0,3) \times (0,3,4) = (0-9,\ 0-8,\ 6-0) = (-9,-8,6),$$

and Π_1 passes through the origin, so

$$\Pi_1 : \quad 9x + 8y - 6z = 0.$$

The plane Π_2

A normal to Π_2 is

$$\mathbf{d}_2 \times \mathbf{d}_2' = (-3,1,3) \times (-4,2,5) = (5-6,\ -12+15,\ -6+4) = (-1,3,-2),$$

and Π_2 passes through $O_2(8a,0,0)$, so $-(x-8a) + 3y - 2z = 0$, that is,

$$\Pi_2 : \quad x - 3y + 2z = 8a.$$

(Quick check: $(-3,1,3)\cdot(-1,3,-2) = 3+3-6 = 0$ and $(-4,2,5)\cdot(-1,3,-2) = 4+6-10 = 0$.)

We assume $a \neq 0$, otherwise the two observers coincide. Then O_2 is not in Π_1 (since $9 \cdot 8a \neq 0$) and O_1 is not in Π_2, which matches what we said above.

The path

The normals $(-9,-8,6)$ and $(-1,3,-2)$ are not parallel, so Π_1 and Π_2 meet in a single line, and that line must be ℓ. Its direction is perpendicular to both normals:

$$(-9,-8,6) \times (-1,3,-2) = (16-18,\ -6-18,\ -27-8) = (-2,-24,-35),$$

so we can take the direction $(2, 24, 35)$. For a point on it, put $y = 0$. The equations become $9x = 6z$ and $x + 2z = 8a$, so $x = \frac{2}{3}z$ and $\frac{8}{3}z = 8a$. This gives $z = 3a$ and $x = 2a$. So $(2a, 0, 3a)$ is on ℓ. It is also on the first sight line from O_1, as it should be.

Therefore the path of P is

$$\boxed{\frac{x - 2a}{2} = \frac{y}{24} = \frac{z - 3a}{35}}$$

or, equivalently, the pair of equations $9x + 8y - 6z = 0$ and $x - 3y + 2z = 8a$.

Finally, ℓ really does meet all four sight lines, at points other than the observers. Writing ℓ as $(2a + 2\lambda,\ 24\lambda,\ 3a + 35\lambda)$, it meets them at

$$(2a, 0, 3a), \qquad (0, -24a, -32a), \qquad \left(\tfrac{80a}{37}, \tfrac{72a}{37}, \tfrac{216a}{37}\right), \qquad \left(\tfrac{56a}{25}, \tfrac{72a}{25}, \tfrac{36a}{5}\right)$$

respectively.

Ask yourself The second point has y and z negative for $a > 0$. Is that a problem?

The sight lines are given as whole lines, and $(0, -24a, -32a)$ satisfies $x = 0$, $y/3 = z/4$. We only used the lines, not which way the observer was facing, so the answer stands. If you read "line of sight" as a ray from the observer, this observation lies behind O_1. That is a quirk of the data, not an error in the method.

Checks

- Substitute the point $(2a, 0, 3a)$ and the direction $(2, 24, 35)$ into both plane equations: $9(2) + 8(24) - 6(35) = 18 + 192 - 210 = 0$ and $2 - 72 + 70 = 0$, so the direction lies in both planes.
- With $a = 1$, the point $\left(\frac{80}{37}, \frac{72}{37}, \frac{216}{37}\right)$ equals $O_2 + \frac{72}{37}(-3, 1, 3)$, and $\frac{80/37 - 2}{2} = \frac{3}{37} = \frac{72/37}{24}$. So it is on both the sight line and the path.
- We checked all four intersection points in exact rational arithmetic for several values of a.

Common slips

- Trying to find where pairs of sight lines meet: lines of sight from different observers at different times need not meet at all.
- Forgetting that Π_2 passes through O_2, not the origin, which gives $x - 3y + 2z = 0$ instead of $= 8a$.
- Sign errors in the cross product. Always dot the normal back with both direction vectors.

Beyond STEP

How many lines meet four given lines in space? For four lines in general position the answer is two (counted over the complex numbers; over the reals it can be zero). That is a classic result of enumerative geometry, and the four lines here are special because they pass through two observers in pairs. The same plane-of-sight idea underlies triangulation and the epipolar geometry of stereo vision, where two cameras recover the positions of points in 3D. For the linear algebra of lines, planes and intersections, read Strang, *Introduction to Linear Algebra*.

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